

# EcoDrive AI

Predictive E20 Ethanol Transition & Engine Health Diagnostics Framework

REAL-TIME OBD-II DIAGNOSTICS

INDIAN FLEET OPTIMIZED



# THE E20 CHALLENGE IN INDIA

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## Mandatory National Shift

As India pushes toward green energy sovereignty, the government mandated E20 Petrol (20% Ethanol blend) as the baseline nationwide grade starting **April 2026**, phasing out E10 entirely.

While models built from 2023 onwards are natively E20-hardened, crores of pre-existing vehicles face unprecedented operational and physical stress.

## Legacy Material Vulnerability

Vehicles built prior to 2023, specifically BS4 and early-BS6 systems, utilize legacy rubber seals and mild steel components never validated for ethanol's corrosive parameters.

This massive, non-backwards-compatible transition risks catastrophic fuel component failure across major Indian transport networks.



# CRITICAL TECHNICAL FAILURE VECTORS

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## Phase Separation

Ethanol is naturally hygroscopic. It actively absorbs moisture from humid ambient air, precipitating water out of gasoline suspension directly to the tank floor, inducing internal galvanic rust.



## Elastomer Crack

Highly volatile ethanol aromatic chains break down the structural binders in traditional rubber hoses and gaskets, making legacy fuel lines dry, brittle, cracked, and prone to high-pressure leaks.



## Lean Burn Conditions

Ethanol has approximately 28% lower gross energy density. Without manual compensation, unoptimized engines burn extremely lean, triggering catastrophic knocking and spike in temperatures.



# ECODRIVE AI PIPELINE ARCHITECTURE

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## Core ML Pipeline

1. 6,000 Sample Vehicle Generation
2. Custom Feature Engineering (TSI)
3. Random Forest Ensembles
4. SHAP Localized Interpretability



## Voice Diagnostics

1. Speech Signal Input Stream
2. "Muesli" Natural Regex Parser
3. Dynamic Parameter Mapping
4. Rules-Engine Diagnostic Match



## Deployment Ecosystem

1. Fast Gradio Interface Web Portal
2. Immediate PDF Document Engine
3. Explanatory SHAP Plot Models
4. Local Maintenance Direct Routing



# AUTOMOTIVE DATASET GENERATION STRATEGY

TABLE 5.1: SYNTHESIZED FLEET REPRESENTATION DISTRIBUTIONS

| Operational Era              | Emissions / Engine Specs                | E20 Vulnerability Index         | Dataset Percentage |
|------------------------------|-----------------------------------------|---------------------------------|--------------------|
| Pre-BS6 Era (<2020)          | Carbureted Fuel Systems, Basic O2 Loop  | Extreme (High Phase Separation) | 25.0%              |
| Transitional BS6 (2020-2023) | Electronic Fuel Injection (EFI) Enabled | Moderate (Elastomer Strain)     | 55.0%              |
| Mandatory E20 (2023+)        | Direct Factory Ethanol Mitigation       | Fully Compatible                | 20.0%              |

\*Physics-informed synthesizer mimics operational conditions using Short-Term Fuel Trim (STFT), Exhaust Gas Temperature (EGT), and misfires.



# FEATURE ENGINEERING: THERMAL STRESS INDEX

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To isolate thermodynamic stress differences between high-capacity car engines and small-displacement commuter motorcycle engines, we engineered the `Thermal_Stress_Index` (TSI) variable:

$$TSI = \frac{EGT}{Engine\_CC + 1}$$

- ✓ Standardizes comparative heat loads in low-displacement (100cc-150cc) Indian commuter platforms.
- ✓ Quantifies the exact temperature drop risk when fuel maps run lean due to the low energy density of ethanol.



# MODEL OPTIMIZATION PHASE

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An optimized Random Forest Classifier utilizes class-weight balancing to resolve demographic fleet skewness. This core ensemble provides high diagnostic explainability for safety-critical systems.

```
# Initialize and Train the Ensemble Classifier Core
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=150,
    max_depth=12,
    random_state=42,
    class_weight='balanced'
)
model.fit(X_train, y_train)
# Evaluate operational predictions
y_pred = model.predict(X_test)
```



# ENSEMBLE VALIDATION PERFORMANCE

TABLE 8.1: PRECISION & RECALL CLASSIFICATION MATRIX

| Target Severity Tier     | Precision      | Recall | F1-Score | Support Fleet |
|--------------------------|----------------|--------|----------|---------------|
| Healthy / E20 Compatible | 0.99           | 0.95   | 0.97     | 626           |
| Degradation Warning      | 0.79           | 0.87   | 0.83     | 205           |
| High Corrosion Risk      | 0.94           | 0.96   | 0.95     | 369           |
| Global Accuracy Core     | 94.0% Accuracy |        |          | 1200          |

Key Validation Metric: 96% recall in high corrosion risk tier guarantees the engine diagnostic loop avoids high-risk false negatives.



# HYBRID DIAGNOSTICS EXPERT ENGINE

## 1. Static Compliance Lookup

Calculates baseline physical resilience values using mechanical manufacturing years and brand-specific material properties.

- ✓ Pre-2020: Structurally Incompatible
- ✓ 2020-2023: Fuel System Material Threat
- ✓ 2023+: Full Compliance Verification

## 2. Dynamic Sensor Rule-Matching

Blends static metadata with active OBD-II parameters to evaluate active risk tiers and deliver custom field mitigation suggestions.

- ⚠ Early warning alerts: Fuel System re-jetting required
- ✗ Immediate crisis action: Phase separation tank flushing



# COMPARATIVE DIAGNOSTIC EXECUTION REPORTS

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## Case A: Legacy Fleet Commuter

**Vehicle:** 2016 Hero Bike (Carbureted)

**Sensors:** STFT 9.2, EGT 740°C, Misfires: 2

**Status:** Incompatible / Critical Risk

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**Mitigation Plan:** Upsize main jets by 10% to eliminate lean burn conditions. Replace standard lines with Viton (FKM) synthetic fuel lines. Run rust preventative coats.

## Case B: Compliant Family Segment

**Vehicle:** 2024 Maruti Suzuki Car (FI)

**Sensors:** STFT 1.1, EGT 645°C, Misfires: 0

**Status:** Fully Compatible / Healthy

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**Mitigation Plan:** Maintain normal scheduled engine maintenance periods. Use high-performance fuel stabilizer additives once per 3 refuels to bind potential moisture.



# ECOSYSTEM SCALE & ROADMAP

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